Enigma Wildcard 2024

Name of the first candidate: _____
Mobile number: **Roll Number:**
Name of the second candidate: _____
Mobile number: **Roll Number:**
Time: 4:00 PM. Date: 15/02/2024
Maximum marks: **120 Marks**

Instructions

- The question paper contains 30 questions containing three types of questions, 1 star (*) {2 mark each}, 2 star (**) {4 mark each}, and 3 star (***) {6 mark each}.
- There is NO NEGATIVE marking.
- Few questions are numeric type, and a few are multiple-choice.
- Every multiple-choice question has four options A, B, C, D, out of which only one is correct and integer value questions have their answer ranging from 1 to 9.
- Mark your answer on the given alphabet option A, B, C, D only.
- For a numeric-type question, write your answer in the space provided beside the question. (Round of to 2 decimal places).
- Log tables, calculators, clipboards, mobile or any other electronic gadgets in any form are not allowed.

- ** There are eight people in a family viz. P,Q,R,S,T,U,V and W consists of 3 generations. Four of them are female. T and R are the daughter and son of Q respectively and both are married. U is the sister of W whose father is S. P and V are of 3rd generation and P is the son-in-law of U. Q is the brother-in-law of W. Who among the following is sister-in- law of T?.

- ** A 3-digit number “x” has 4 distinct prime factors. The difference of pair of two factors multiplied together is 11. The sum of digits in the given number is 3. Find 2x.

- *GREECE: @$@.@*

  GERMANY: *+$@#\{
  RUSSIA: (-*(+/
  MANGO: ?

  A. #\$+{
  B. @\+$(
  C. #\+$)
  D. \#+*)

- *.M, N, O, P, Q and R are two sets of triplets, not necessarily in the order given.

  i. Four of the six are male and two are female.
  ii. Neither set of triplets contains 3 siblings of the same sex.
  iii. M and P are married to each other.
  iv. N is Q's only brother.
  Who cannot be sister and brother?

  a)M and Q
  b)O and R
  c)R and P
  d)R and Q

- *Chef has made a delicious Cake of Regular Pentagon Shaped with endpoints of the Cake named as A, B, C, D, E respectively. O is a point is a point inside the cake such that AOB is an equilateral triangle. What is ∠OEA?

  1)66 2) 72 3)54 4)48

- **The reflex angle between the hands of a clock at 10.25 is:

  1.180°
  2.192.5°
  3.195°
  4.197.5°

- *CREDENZ: XIVWVMA

  PISB: ?
  a. LRHZ
  b. KRIY
  c. KRHY
  d. LQIY

- ***Place 6 queens on the given 6x6 board such that no queen attacks other queen.

- **If A#B means A is the mother of B .A-B means A is the brother of B. A%B means A is the father of B. A*B means A is the sister of B. Which of the following is the following shows that P is the maternal uncle of Q?

  a. Q-N#M*P
  b. P#S*N-Q
  c. P-M#N*Q
  d. Q-S%P

- *Mary is walking towards South-West for 15 miles, turns right and walks another 10 miles. She then turns 90 degrees clockwise and walks 10 miles. Now, she turns 90 degrees anticlockwise and walks 30 miles. Which direction is she facing?

  a. East
  b. West
  c. South-East
  d. North-West

- *In certain language, TEARS is coded as 18, MOUNTAIN is coded as 35, then how will be WATER coded?

  a.25
  b.22
  c.18
  d.20

- **6 boys Ramesh, Suresh, Dipesh, Rupesh, Paresh, Kamlesh and 6 girls Seema, Geeta, Rita, Meena, Komal, Radha are standing in rows, such that each girl faces one boy, not necessarily in the same order. Seema is on the immediate right of the girl who is facing Paresh, the boy at extreme right. Only Suresh is between Rupesh and Paresh. Kamlesh is to the immediate left of Ramesh and to the immediate right of Dipesh. Rita is facing Ramesh and is to immediate left of Geeta. Radha is third to the left of Meena. Which girl is facing Rupesh?

a. Geeta
b. Seema
c. Radha
d. Meena

- **P can do a piece of work in 5 days of 8 hours each and Q can do in 4 days of 6 hours each. How long will they take do it working 5 hours a day?

a. 2 days
b. 3 days
c. 4 days
d. 5 days

- **The year after1896 that will have the same calendar as that of the year 1896 :

a. 1902
b. 1904
c. 1905
d. 1908

- **This question is based upon the information given below. Study the information carefully and then choose the correct alternative to answer the question. Five friends A, B, C, D and E are sitting on a bench. (1) A is sitting next to B. (2) C is sitting next to D. (3) D is not sitting with E. (4) E is on the left end of the bench. (5) C is on the second position from right. (6) A is on the right side of B and to the right side of E. (7) A and C are sitting together. Draw the arrangement.

- **In the question below, two statements are given followed by two conclusions. Take the given statement to be true despite being at variance with known facts. Find which of the given conclusion(s) logically follow(s) from the given statements. Statements: All doraemons are nobitas. Some nobitas are jiyans. Conclusions: I- Some doraemons are jiyans II- Some jiyans are nobitas (1) Only I follows (2) Only II follows (3) Either I or II follows (4) None follows

- **Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10

and 12 seconds, respectively. In 30 minutes, how many times do they toll together?

- **For which of the following will the expression

  $(x^3 + 121)(x^2-81)(x + 49)$ be odd? (multiple answers may be correct)

  a. x + 3 is even.
  b. 3x is even.
  c. x/3 is an even integer.
  d. 7x-9 is even.
  e. 11x + 5 is odd.

- ***Three pendulums start oscillating together at the same time. The first oscillates 2 times every 5 seconds, the second 3 times every 7 seconds and the third 1 time every 4 seconds. What is the smallest time interval when all three will oscillate together again?

- ** Harry set the alarm on his old clock for Thursday at 10.30 am. He set his clock correctly on Wednesday 12.30 am. He was unaware that the clock gains 30 min every 51 hrs. When his alarm went off, what was the actual time?

A .10:50
B. 10:40
C. 10:58
D. 10:10

- **A digital lock has 3-digit key

The hints are-----
1) (682) One number is correct and well placed.
2) (614) One number is correct but wrongly placed.
3) (206) Two numbers are correct and both wrongly placed.

- **1 1 1 1 = R

2 2 2 2 = T
3 3 3 3 = E
4 4 4 4 = ?

- * m>0,

$m^{13/15} = 41/71$
Compare m....41/71

- *** I am the only number whose sum of digits is equal to the index of mine in an array of all the atmost 2-digit prime numbers when indexed from right to left. (The indexing starts from 1). Who am I?

- ***You have 12 coins amongst which 1 is counterfeit. The counterfeit coin can either be heavier or lighter. What is the minimum number of times you'll have to use a balance scale to find the counterfeit coin?

- ***You have a magical bag. The greater the number of chocolates that you put in the bag the more comes out. If you use the magical bag twice, the amount that comes out is three times the amount that you started with. How many chocolates will come out if you use the magical bag only once on 14 chocolates?

- **You and your mathematician friend are trapped in a dungeon. Your friend manages to escape before you. Your mathematician friend the following hints to crack the three-digit password:

a. The product of the three digits is 36.
b. The digits are in ascending order.
c. The sum of the three digits is the hallway number your friend entered from.
d. The largest digit appears only once.
Find the number.

- **Place the numbers 1, 2, 3, 4, 5, 6, 7, 8 into the eight circles in figure given below, in such a way that no number is adjacent to a number that is next to it in the sequence. For example, 1 should not be adjacent to 2 but can be adjacent to 3, 4, 5, 6, 7, 8. Similarly for others.

- ***Given a Circle and Six Straight Lines. What is the maximum number of pieces that one can cut the circle using these six straight lines?

- ***There are 4 persons (A, B, C and D) who want to cross a bridge in night.

A takes 1 minute to cross the bridge.
B takes 2 minutes to cross the bridge.
C takes 5 minutes to cross the bridge.
D takes 8 minutes to cross the bridge.
There is only one torch with them, and the bridge cannot be crossed without the torch. There cannot be more than two people on the bridge at any time, and when two people cross the bridge together, they must move at the slower person's pace.
What is the minimum time required for all of them to cross the bridge?